International Journal of Sports Physiology and Performance, (Ahead of Print)
https://doi.org/10.1123/ijspp.2023-0013
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First Published Online: Apr. 28, 2023

Human Kinetics
ORIGINAL INVESTIGATION

Programming High-Speed and Sprint Running Exposure in Football: Beliefs and Practices of More Than 100 Practitioners Worldwide

Antonio Dello Iacono,[1] Marco Beato,[2] Viswanath B. Unnithan,[1] and Tzlil Shushan[3]

[1]School of Health and Life Sciences, Institute for Clinical Exercise and Health Science, University of the West of Scotland, Hamilton, United Kingdom; [2]School of Health and Sports Sciences, University of Suffolk, Ipswich, United Kingdom; [3]School of Health Sciences, Western Sydney University, Sydney, NSW, Australia

Purpose: To gain knowledge on the beliefs and practices of football practitioners applying high-speed and sprint running exposure programming strategies. ***Methods:*** One hundred two football practitioners from 22 different countries participated in a study consisting of a survey including 5 domains: demographic and professional characteristics (*Who*); importance of high-speed and sprint running exposure for physical-capability development, preparation for competition, and injury-prevention strategies (*Why*); exposure timing (*When*); methodological procedures for exposure monitoring and training scheduling (*What*); and effectiveness of common training practices (*How*). Data were analyzed using a combination of descriptive statistics, generalized mixed effects, and multinomial logistic regression models. ***Results:*** Data revealed 5 main findings: (1) overall agreement on the importance of exposure for physical-capability development, preparation for competition, and injury-prevention strategies; (2) different exposure timing and selective training scheduling for starting and nonstarting players across typical and congested weeks; (3) lack of consensus on the conceptual constructs defining high-speed and sprint running metrics and the methodological procedures used for monitoring; (4) a probable association between match-related outcomes and exposure strategies used in training; and (5) a broad range of training methods considered effective to elicit exposure. ***Conclusions:*** This study provides actionable insights into the planning, implementing, and monitoring strategies for high-speed and sprint running exposure in football. While some conform with the evidence on high-speed and sprint running training in football, further research and professional debate are warranted to develop empirical knowledge and provide pragmatic recommendations to help practitioners in adopting evidence-informed decisions.

Keywords: external-load monitoring, strength and conditioning, training load

Training-load programming and its constituent components (ie, planning, implementing, and monitoring) are paramount in management strategies aiming to optimize football (soccer) training.[1] In fact, compelling evidence exists on the relationships between training load and (1) physical development, (2) football performance, and (3) injury risk among football players of any sex, age, and competitive level.[2–4]

Programming training load has become a more challenging task in modern football due to the high physical demands associated with the evolving locomotor profile of this sport as well as with the increasing number of matches played across repeated congested fixture periods during a season.[5,6] Of particular interest is the ongoing debate among football practitioners about the most appropriate training load programming strategies pertaining to high-speed (HSR) and sprint running (SR) exposure.[7,8] While HSR and SR exposure seems to be determinant of physical preparation,[3,4] football performance,[9–13] and injury mitigation strategies,[14,15] evidence-based recommendations are somewhat lacking and much remains unknown in this domain.

With recent developments of tracking technology, accurate and reliable HSR and SR exposure monitoring is now a widespread routine in applied football settings.[16] In a recent study describing the practices to develop sprint performance in elite football code athletes, Nicholson et al[17] identified that practitioners use integrated monitoring approaches to inform sprint training prescription as well as a combination of training strategies, methods, and protocols for sprint development. Nonetheless, little is known of the beliefs and practices of football practitioners applying specific HSR and SR exposure programming strategies while facing the contextual day-to-day challenges. Gaining knowledge of beliefs and common practices in this domain can provide actionable insights for decision-making processes, serve as basis for further professional debate, and inform future applied football research. Therefore, the aim of this study was to survey: (1) the beliefs of practitioners about the importance of HSR and SR exposure in football, and (2) the associated common practices and methods for their monitoring and training prescription and implementation.

Methods

Subjects

A convenience sample of football practitioners were recruited via email, personal, or group messaging applications (eg, WhatsApp), and promoted on social media (eg, Facebook and Twitter) through the professional networks of the research team. Eligibility criteria

Beato https://orcid.org/0000-0001-5373-2211
Unnithan https://orcid.org/0000-0001-5147-1679
Shushan https://orcid.org/0000-0002-0544-1986
Dello Iacono (antonio.delloiacono@uws.ac.uk) is corresponding author, https://orcid.org/0000-0003-0204-0957

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were the following: be ≥18 years old; be a football practitioner (eg, any member of the coaching, performance, support, or medical staff); currently work or have worked in elite, professional, or semiprofessional level; have experience with HSR and SR exposure programming (ie, planning and implementing monitoring) practices. Participants provided informed consent, and the study received university ethical approvals (RETH[S]21/014).

Design

A cross-sectional, survey study design was used to survey practitioners' beliefs and practices pertaining to HSR and SR exposure programming in football. The survey (Supplementary File 1 [available online]: https://osf.io/8dfbs) was designed in English language using an online platform (Qualtrics[XM] https://www.qualtrics.com/au). Questions were developed based upon domain expertise of the authors as well as in consultation with academic peers and football practitioners. Pilot surveys (n = 5, Supplementary File 2 [available online]: https://osf.io/s6we9) were tested to achieve agreement among the authors prior to the release of the final version, and data were collected between March 9 and July 5, 2022.

Survey Content

As illustrated in Figure 1, the survey included 5 domains:

- "*Who*" — Participants' details including demographic data and professional characteristics of the participants and their working environment.
- "*Why*" — An array of 5-point Likert scales (from "strongly disagree" to "strongly agree") questions on the perceived importance of HSR and SR exposure for: (1) development of physical capabilities, (2) preparation for competition, and (3) injury prevention strategies.
- "*When*" — A combination of ranked and multiple checkbox options to compare exposure timing: (1) match versus training, (2) typical (ie, 2 matches 1 wk apart) versus congested fixture weeks (ie, 3 matches across the week), and (3) starting versus nonstarting players.
- "*What*" — An array of 1 to 2 loops of combined questions (checkbox, numerical values, and open-ended text) were used to gain understanding on the methodological procedures implemented for monitoring exposure and scheduling training.
- "*How*" — A 5-point Likert scale (from "not effective at all" to "extremely effective") examining agreement levels on the effectiveness of common training practices for exposure training.

To avoid ambiguity around definitions and questions, interpretation and written examples were provided throughout the survey. Furthermore, participants were given the option to elaborate on their responses or provide more details using open-ended text.

Data Handling

Data from questions with preset answers (ie, predefined single or multiple choices) were converted into standardized codes using a designated Microsoft Excel spreadsheet; all automated responses were checked for veracity. The remaining data (ie, open-ended answers) were analyzed independently by 2 authors (Shushan and Dello Iacono) using the same standardized codes. Relevant information was added or discarded through a discussion between the same 2 authors, while a third author (Beato) acted as moderator in a case of disagreement. The full data set is available as Supplementary File 3 (available online): https://osf.io/qde2t.

Statistical Analysis

All statistical analyses and visualizations were conducted in R language and environment for statistical computing using the *ggeffects, lme4, nnet, sjPlot,* and *tidyverse* packages while model assumptions were checked using the *DHARMa* package (version 4.2.1).

Due to the cross-sectional and observational study design, data are presented using a variety of descriptive statistics. Single and multiple-choice questions are reported using mean and standard deviation (mean [SD]), median and interquartile range, mode, range, and frequencies (absolute [counts] and relative [percentage]).

Regarding the "*When*" domain, we compared HSR and SR exposure scheduling between starting and nonstarting players using a generalized mixed-effects model. Since *players status* (starting vs nonstarting) was treated as binary outcome variable, a binomial error distribution was specified with a logit-link function to predict the odds associated with the predictor variable *exposure timing* (categorical variable with 3 levels [match, full-squad training, and complementary training]). Two contrasts were set to examine: (1) odds between match versus the 2 training options pooled together and (2) odds between full-squad training and compensatory training. Moreover, random effects were assumed for participants. However, upon generating the model outputs' summary, we noted that the within-subject variance explained by the random effects was 0. Therefore, we opted to use a more parsimonious generalized model by retaining *exposure timing* as a predictor and removing the random effect for participants. Akaike information criterion score was examined to confirm the selection of the final model to obtain the best-fit model while maintaining model parsimony. Odds ratios are presented to aid interpretation of the findings. To validate the assumptions of the generalized mixed-effects model, tests for uniformity of residuals, under- and overdispersion, outliers, and zero inflation were performed using a simulation-based approach, which confirmed the absence of significant violations of the model fit.[18]

We used multinomial logistic regression models to investigate how exposure target determined from match-related outcomes (ie, percentage of total match exposure) affects HSR and SR training strategies in typical weeks. First, we considered match outcomes as continuous variables in view of their normal distribution observed during the exploratory data analysis step. To this end, the responses (n = 5 and n = 4 for HSR and SR exposure subsets, respectively) corresponding to "*match outcomes >200%*" were removed from the original data sets upon verifying that their removal improved the predictive accuracy and overall fit of the model without affecting the point estimates. We provide the code underpinning the procedural steps leading to this decision in Supplementary File 4 (available online): https://osf.io/erv3p. Since *exposure strategy* was treated as a categorical outcome variable with 3 levels (single session, 2 sessions with microdoses, and multiple sessions with microdoses), 2 binomial error distributions were specified with logit-link functions to predict the relative odds (2 sessions with microdoses or multiple sessions with microdoses vs single session) associated with the continuous predictor variable *match load outcomes* (percentage). For microdosing, we refer to the practice of splitting up the total weekly external training load exposure into multiple (2 or more) sessions spaced out across the week.[19] For the interpretation of the outputs, we avoid using a dichotomous approach based upon

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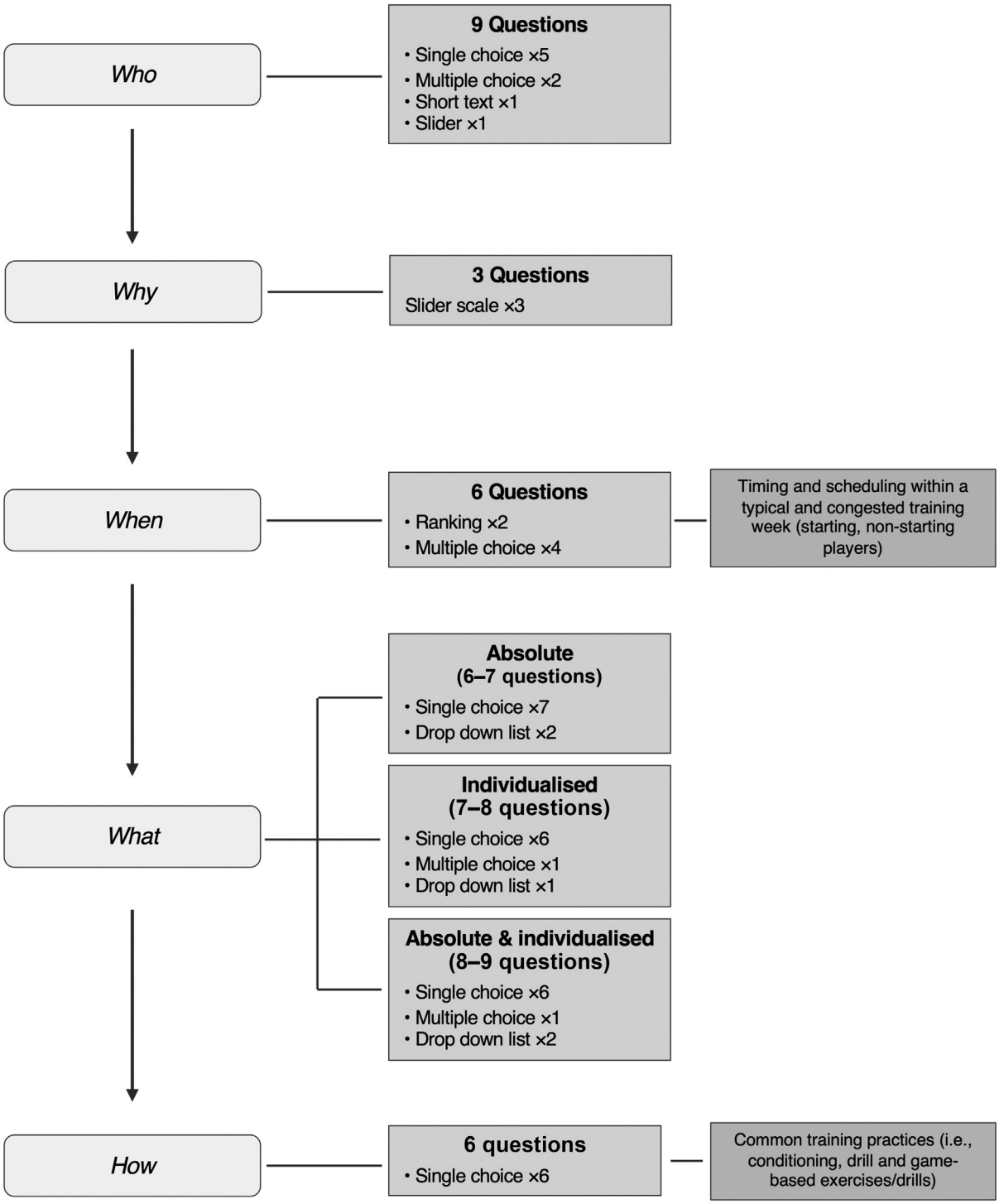

Figure 1 — A flowchart presenting the survey's domains and question characteristics.

traditional null hypothesis significance testing, which has been extensively criticized.[20] Alternatively, in the discussion section, we contextualize the practical implications of the results by providing examples with an emphasis on the predicted probabilities to aid interpretation of the findings. To validate the linear multinomial logistic regression models, the assumptions of independence of irrelevant alternatives, linear relationships between log-odds of the outcome, and independent variable and independence of errors were confirmed.

Results

The "Who"

One hundred two football practitioners (female: n = 1; male: n = 100, unknown: n = 1) with 9.3 (9.1) years (range: 1–45) of experience volunteered to participate in this study. Respondents' characteristics and working environment details are presented in Tables 1 and 2, respectively.

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Table 1 Practitioners' Demographic, Role, and Education Characteristics

Country	n	%	Current role	n	%	Highest qualification	n	%
Italy	42	41.1	Strength and conditioning	56	54.9	Vocational	6	5.9
England	14	13.7	Sport scientist	24	23.5	Undergraduate	24	23.5
Belgium	9	8.8	High-performance manager	5	4.9	Postgraduate	50	49.0
Israel	7	6.9	Technical/tactical coach	3	2.9	Doctorate	16	15.7
Australia	5	4.9	Football coaching staff	3	2.9	Other	6	5.9
Spain	3	2.9	Physical therapist/physio	2	1.9			
Germany	3	2.9	Other	9	8.8			
United States	3	2.9						
Scotland	2	2						
Poland	2	2						
France	1	1						
Denmark	1	1						
Finland	1	1						
Norway	1	1						
Malta	2	2						
Portugal	1	1						
The Netherlands	1	1						
Canada	1	1						
Macedonia	1	1						
Switzerland	1	1						
Greece	1	1						

Table 2 Players Profile

Sex	n	%	Age	n	%	Level	n	%	Number of pitch sessions	n	%
			Multiple age groups (N = 20)			Multiple levels (N = 10)					
Male	85	83	Youth	11	10.8	First tier	60	53.6	2–3	16	15.7
Female	7	6.9	Academy	34	33	Second tier	23	20.5	4–5	51	50.0
Both	10	9.8	Senior	77	77	Third tier	13	11.6	>5	35	31.3
						Fourth tier	10	8.9			
						Fifth tier	6	5.4			
						Sixth tier	1	0.9			
						Other	1	0.9			

The "Why"

The perceived importance of exposure to HSR and SR was comparable across the 3 training load domains (Figure 2).

The "When"

Practitioners reported the greatest exposure (ranked first) for starting and nonstarting players during official matches (n = 77) and complementary training sessions (n = 61), respectively. Full-squad training was ranked second by most practitioners for both groups (Figure 3). The generalized model revealed a main effect of exposure timing on player status (Table 3). Figure 4 illustrates the scheduling strategies during typical and congested weeks. For starting players (panel A), exposure was more commonly scheduled on match day (MD) − 3 (44%) during typical weeks, with no training exposures scheduled in congested weeks (60%). Exposure among nonstarting players (panel B) was more commonly scheduled "*immediately postmatch*" (24%), on MD + 1 (21%) and M − 2 (22%) during typical weeks, and "*immediately postmatch*" (38%) in congested weeks.

The "What"

Figure 5 presents a summary of the methodological procedures used for HSR and SR monitoring. Generally, practitioners used absolute thresholds (n = 46 and 38 for HSR and SR, respectively), often in combination with individualized speed thresholds (n = 32 and 37). Maximum speed and maximal aerobic speed were the most common anchors among practitioners using individualized thresholds alone (n = 24 and 27 for HSR and SR, respectively).

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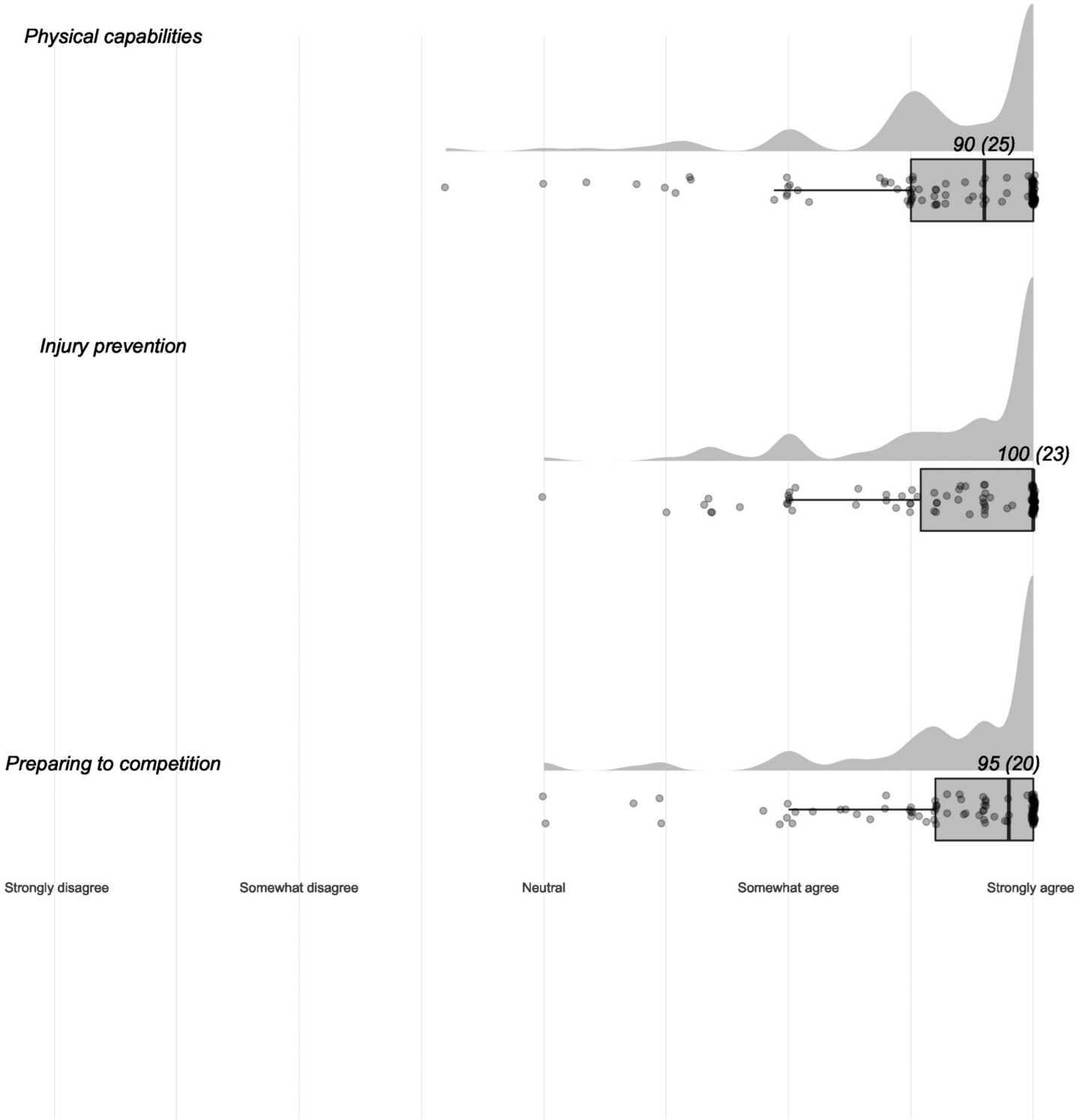

Figure 2 — A raincloud plot presenting practitioners' perceived importance regarding the exposure to high-speed and sprint running for physical-capability development, preparation for competition, and injury-prevention strategies. Labels represent the median and interquartile range.

As shown in Figure 6, most practitioners reported using match-related outcomes as a reference for HSR (n = 68) and SR exposure (n = 72) prescription during the training microcycle. However, there was a large between-practitioner variability regarding the exposure target (ie, percentage of total match exposure). Nearly half (n = 49) reported prescribing 2 sessions with microdoses, with the remaining prescribing a single session (n = 33) or multiple sessions with microdoses (n = 18).

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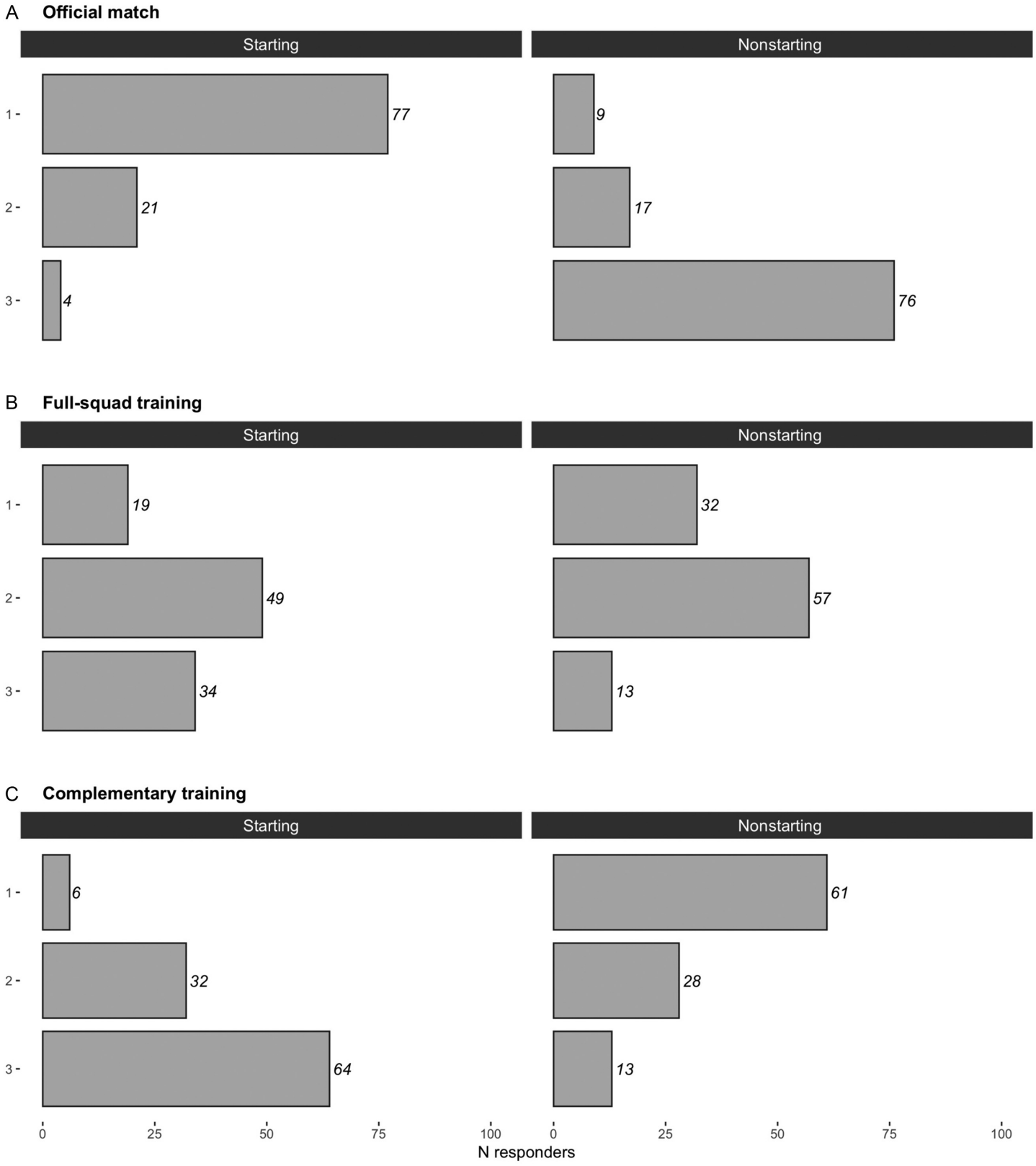

Figure 3 — Frequencies of ranking responses regarding the greatest to the least exposure of high-speed and sprint running during an official match (panel A), full-squad training (panel B), and complementary training (panel C) among starting and nonstarting players.

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Table 3 Logistic-Regression Outputs of High-Speed and Sprint Running Exposure Timing

	Odds ratio	CI	*P*
Match	0.89	0.59–1.36	.573
Full-squad training and complementary training vs match	2.74	2.09–3.72	<.001
Complementary training vs full-squad training	4.10	2.55–7.10	<.001

Overall, the multinomial logistic model revealed a probable association between match outcomes and exposure strategy. Specifically, a greater exposure target was associated with an increased relative probability of implementing 2 sessions with microdoses and a concurrent reduced relative probability for either a single session or multiple sessions with microdoses both for HSR and SR training. The model outputs are reported in Table 4 with the associated predicted probabilities displayed in Figures 7 and 8.

The "How"

The perceived effectiveness of common training methods to elicit HSR and SR exposure is displayed in Figure 9.

Discussion

This study provides insights into beliefs and practices of football practitioners regarding programming strategies for HSR and SR exposure. The main findings were the following: (1) overall agreement on the importance of exposure for physical capabilities' development, preparation for competition, and injury prevention strategies; (2) different exposure timing and selective training scheduling for starting and nonstarting players across typical and congested weeks; (3) lack of consensus on the conceptual constructs defining HSR and SR metrics and the methodological procedures used for their monitoring; (4) a probable association between match-related outcomes and exposure strategies used in training; and (5) a broad range of training methods considered as effective to elicit exposure.

The "Why"

We observed consistent agreement between practitioners from several countries on the importance of exposure across the 3 domains of training load management strategies. In fact, most respondents at least "*somewhat agreed*" that exposure has an important role for physical capabilities' development (n = 93), preparation for competition (n = 95), and injury prevention strategies (n = 95). These findings are not surprising considering the evidence on the athletic surrogates to football performance and the locomotive demands of match play, which require players to perform HSR and SR actions repeatedly during a match to fulfill positional-specific tactical responsibilities.[5,9,10,21] HSR and SR activities are also the most frequent locomotive actions preceding goal situations, performed by either the scoring player or the assisting one.[11,12] Therefore, their perception as key training contents for competition preparation seems appropriate. Similar findings were also observed in the study of Nicholson et al,[17] in which practitioners reported that the main rationale for sprint development training prescription was targeting the sport-specific locomotive demands and inducing physical adaptations underpinning a positive transfer onto football performance.[17] Finally, as injury prevention is an established priority of training load management strategies, with injury mechanism studies highlighting the harmful association between unaccustomed volumes and spikes in sprint and near-to-maximal speed distances and muscle injury occurrence,[14,15,22,23] it seems logical that practitioners strongly agreed on the importance of exposure in this respect.

The "When"

The responses pertaining to the scheduling domain revealed actionable insights. The odds to expose nonstarting players to HSR and SR were nearly 3 times greater in training than matches compared with starting players (Table 3). This finding aligns with the evidence reporting the strong association between playing status and HSR and SR exposure.[24,25] In fact, the absent or partial match-induced exposure among nonstarting players requires compensatory strategies that can be actioned only in training. Interestingly, further analyzing the timing domain data indicated that full-squad training was not considered the elective option to ensure the greatest exposure among nonstarting players. Practitioners ranked complementary training as the most appropriate alternative, with a probability to induce the largest exposure even 4 times greater than full-squad training (Table 3). From an operational perspective, this implies that different exposure strategies are required between starting and nonstarting players within the same microcycle. In typical weeks, when consecutive matches are played 7 to 8 days apart, this programming task is reasonably manageable. However, its complexity increases considerably in congested fixture weeks due to the multifactorial demands arising from the interactions between players' status and scheduling constraints. In fact, previous and upcoming matches are also key factors to consider when planning adjustments to exposure within a microcycle. In this scenario, practitioners should consider implementing synchronous, but distinct, exposure strategies as an optimal solution addressing the intricacies between the training load management demands and constraints described above.[25] Data from this survey study reflect this assumption. To illustrate, 99 practitioners indicated that the preferred scheduling options for starting players in typical weeks would be any days from at least 48 hours post the previous match (ie, MD + 2) to not later than 48 hours prior to the upcoming match (ie, MD − 2). In congested weeks, while most practitioners (n = 61) considered exposure unnecessary for starting players, still 30 of them indicated MD − 2 (in reference to the upcoming second and third matches) as the most appropriate option. These findings are likely reflective of the robust evidence on the recovery kinetics of inflammation status, muscle damage, perceptual responses, and physical performance, which require between 48 and 72 hours to return to baseline levels after football matches.[26] While the same reasons may explain part of the responses on scheduling strategies for nonstarting players, we observed large between-practitioner

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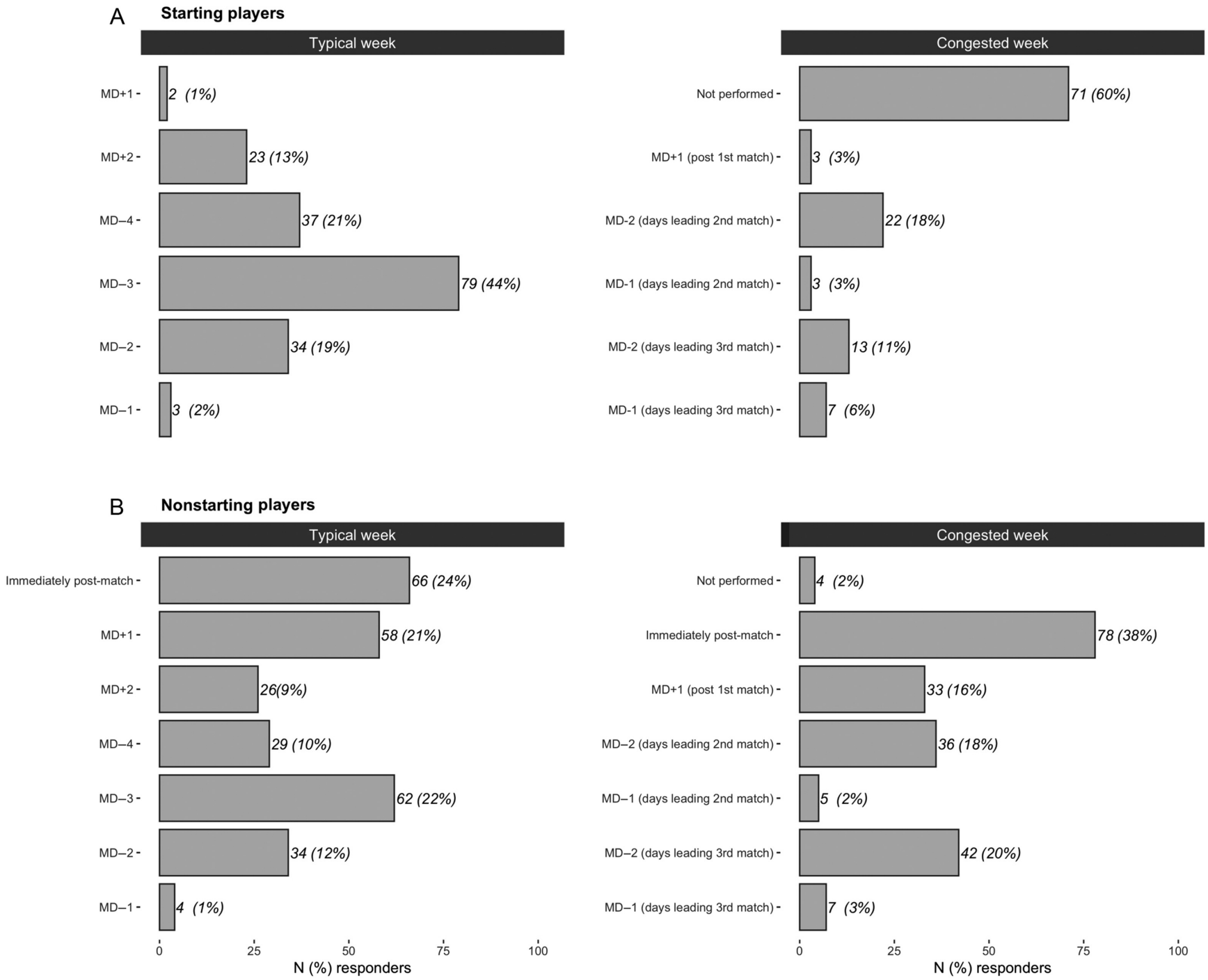

Figure 4 — Frequencies of high-speed and sprint running scheduling strategies among starting and nonstarting players while considering typical and congested training weeks.

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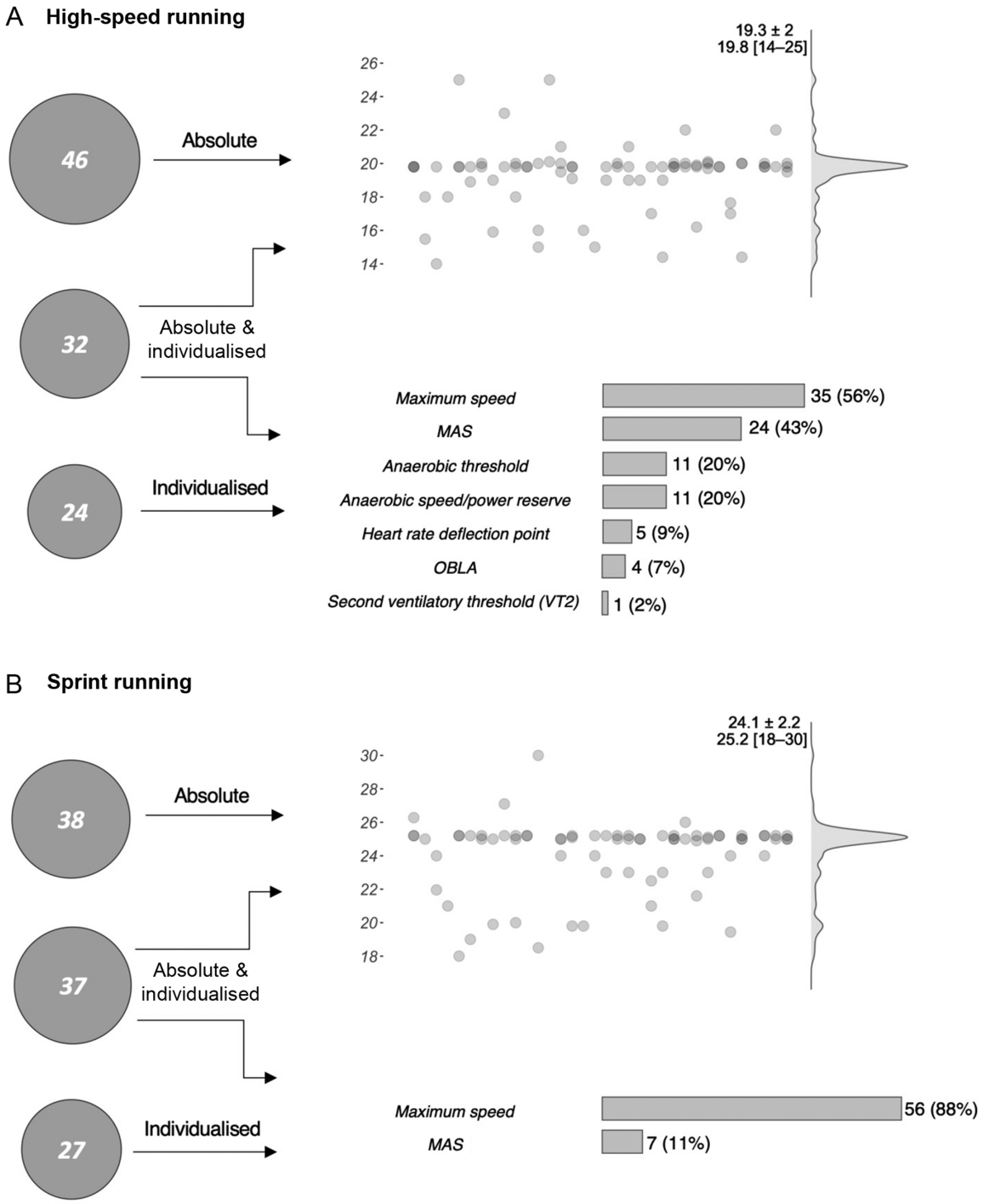

Figure 5 — Practitioners' methodological procedures for obtaining high-speed and sprint running velocity thresholds. Numbers represent mean ± SD, and mode [range]. MAS indicates maximum aerobic speed; OBLA, onset of blood lactate accumulation; VT2, second ventilatory threshold.

variability regarding the most appropriate timing both in typical and congested weeks. To explain, while 54 participants selected a mixture of days between MD + 2 and MD − 2 as the best scheduling option in typical weeks, the remaining 46 participants suggested "*immediately postmatch*" and MD + 1 as preferred alternatives. An opposite balance was observed for congested weeks, with "*immediately postmatch*" and MD + 1 indicated by practitioners and the 2 MD − 2 relative to the upcoming matches indicated by 39 practitioners. Although further qualitative details were not collected in this study, we speculate that other contextual factors may influence scheduling decisions for nonstarting players such as traveling schedule constraints especially when playing away with late evening kick-off time, facilities and training ground availability for compensatory sessions immediately postmatch, relative importance placed on previous and upcoming matches, and the associated rotation strategies between starting and nonstarting players.

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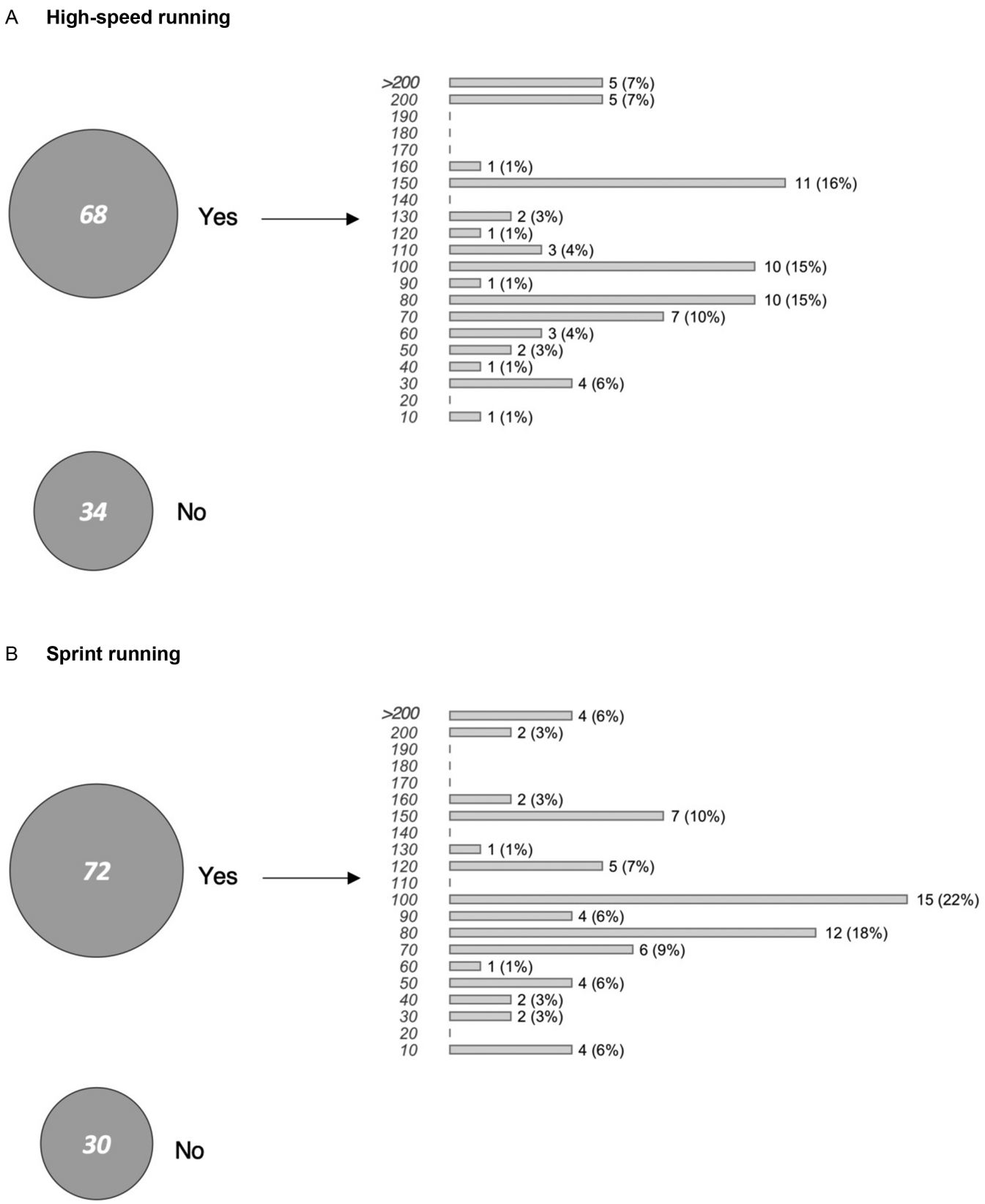

Figure 6 — Frequencies of the use/nonuse of match-related outcomes as a reference for high-speed (panel A) and sprint (panel B), as well as their exposure targets (ie, percentage of total match exposure) during a typical training week.

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Table 4 Multinomial Regression Outputs of High-Speed and Sprint Running Exposure Training

Predictor	Response	Odds ratio	CI	*P*
High-speed running exposure				
Single session	Two sessions with microdoses	0.48	0.10–2.24	.342
	Multiple sessions with microdoses	1.85	0.32–10.57	.485
Match outcomes, %	Two sessions with microdoses	1.01	1.00–1.03	.108
	Multiple sessions with microdoses	0.99	0.97–1.01	.227
Sprint running exposure				
Single session	Two sessions with microdoses	1.12	0.27–4.67	.870
	Multiple sessions with microdoses	1.80	0.33–9.85	.495
Match outcomes, %	Two sessions with microdoses	1.00	0.99–1.02	.495
	Multiple sessions with microdoses	0.99	0.97–1.01	.155

Note: % indicates percent points relative to match outcomes.

The "What"

The large heterogeneity between respondents regarding velocity thresholds is not surprising and likely arises from the different arbitrary thresholds between software manufacturers and methodological choices related to player's age, performance level, or sex.[27] Practitioners reported utilizing various methods (often in combination) to establish individualized HSR and SR thresholds. While the reasons underpinning the use of a particular or mixed methods cannot be ascertained from the current data, they may derive from human and technology resources' availability, the utilization of approaches across different stages of the season (eg, preseason vs in-season), or simply reflect the lack of consensus and conceptual constructs for individualized velocity thresholds in football.[7,28,29] For example, the use of multiple physiological markers (ie, running speed corresponding to the respiratory compensation threshold or maximal oxygen consumption) and performance measures (ie, maximal aerobic speed or maximal sprint speed) characterizing the transitions between intensity domains would enhance the understanding and interpretation of the individual's training and match demands.[30] Similarly, an individualized approach to HSR and SR exposure monitoring may facilitate appropriate recovery and periodization schedules to manage loads and optimize adaptation to training.

A deeper analysis of how using match-related outcomes as exposure target influences training strategies for starting players during typical weeks revealed a common trend. A greater exposure target was associated with an increased relative probability of implementing 2 sessions with microdoses and concurrent reduced probabilities for a single session or multiple sessions with microdoses. To contextualize, when the exact match outcomes (ie, 100%) were used as target, the probabilities of implementing single, 2, or multiple sessions were 33%, 48%, and 19%, respectively, for HSR and 32%, 53%, and 15%, respectively, for SR. However, a target exposure 2-fold (ie, 200%) greater than match outcomes was associated with 18%, 79%, and 3% probabilities for single, 2, or multiple sessions, respectively, for HSR and 29%, 68%, and 3% for SR (Figures 7 and 8). While these findings should be interpreted with caution in view of the uncertainty of the predictive model (eg, overlap of the confidence intervals), it is highly probable that the preferred strategy consists of 2 sessions with microdoses especially when the reference target is greater than match-related outcomes (ie, >100%). It is legitimate to assume that the rationale for this choice is to mitigate the risk that unaccustomed cumulative (eg, from different drills within a session) HSR and SR training loads might produce adverse effects (eg, excessive metabolic stress, muscular damage, and fatigue), thus requiring appropriate exposure strategies to facilitate optimal recovery and adaptations. Spacing out the target weekly HSR and SR exposure across multiple sessions or bouts may still induce stimuli that effectively develop or maintain key physical capacities (eg, acceleration, maximal speed, aerobic, and anaerobic power) while limiting excessive fatigue especially in proximity to match day.[19,31] Applying these findings in the context of the scheduling domain, a practical suggestion would be to plan 2 sessions between MD + 2 and MD − 2, possibly ~48 hours apart and with the cumulative exposure either equally or unevenly split according to the preferred periodization approach. This solution reflects externally valid evidence on football microcycle periodization and represents a reasonable compromise for full-squad training when selective strategies between starting and nonstarting players are not feasible or practical.[32,33]

The "How"

The responses on the effectiveness of common football training methods for HSR and SR exposure delineated what seemed to be a large consensus. Most practitioners indicated that conditioning (n = 85 and n = 75 for HSR and SR, respectively) and drill-based exercises (n = 63 and n = 54 for HSR and SR, respectively) were at least "*very effective*" to ensure adequate exposure (Figure 9). These findings were somewhat expected as training prescribed in the form of high-intensity or maximal-speed predetermined and fixed runs, executed without the ball or replicating football-specific paths with ball involvement that allows to dictate and control the locomotive pace, whereby inducing the target exposure.[7,34–36] Surprisingly, game-based exercises were largely considered at least as "*moderately effective*" both for HSR (n = 90) and SR (n = 76) exposure. While a quantitative interpretation of the practitioners' perceptions was not possible as responses could not be converted into metrics of HSR and SR exposure, this finding deviates from the literature on football sided-games training and the results of a recent systematic review and meta-analysis of our group on this topic.[28] Briefly, we found that sided-games training is inappropriate to induce HSR and SR exposure irrespective of the format characteristics and playing constraints, unless very extensive training

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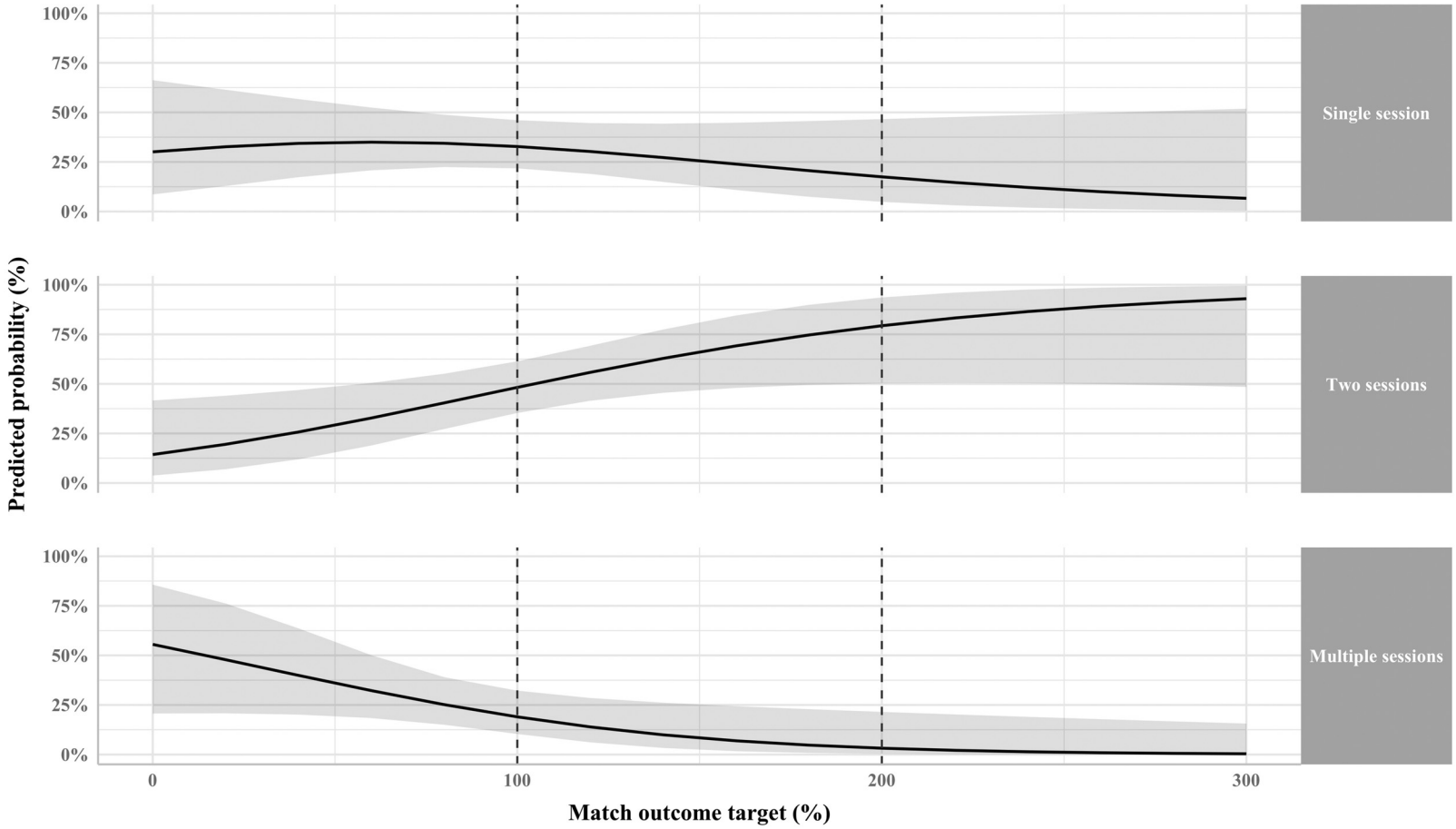

Figure 7 — Predicted probabilities of high-speed running exposure training strategies. Dotted vertical lines represent exposure targets equal to match outcomes (ie, 100%) or 2 times the match outcomes (ie, 200%).

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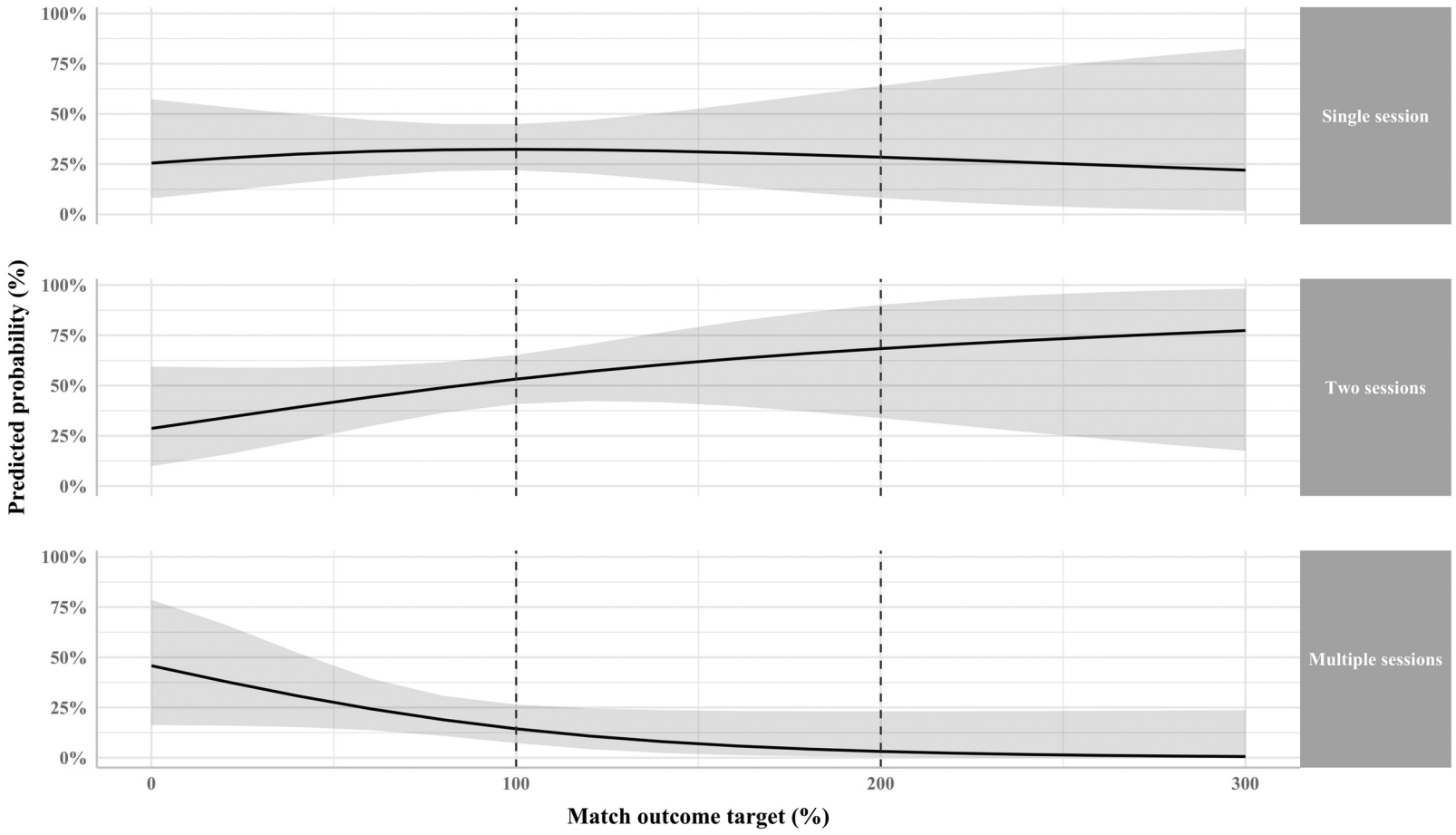

Figure 8 — Predicted probabilities of sprint running exposure training strategies. Dotted vertical lines represent exposure targets equal to match outcomes (ie, 100%) or 2 times the match outcomes (ie, 200%).

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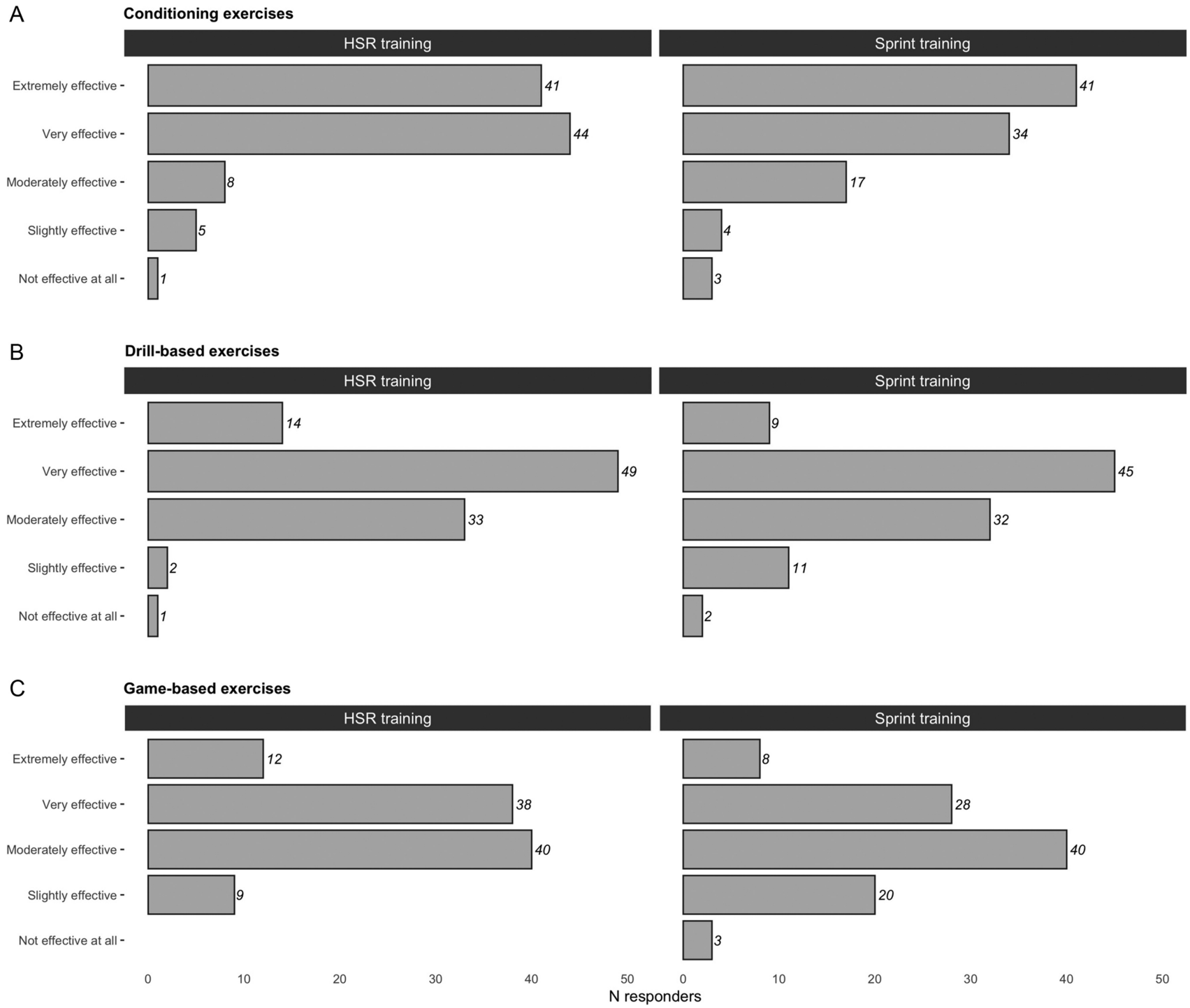

Figure 9 — Perceived effectiveness of training methods to elicit HSR and sprint running exposure. *Note.* HSR indicates high-speed running.

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volumes and formats including small numbers of players (ie, from 2v2 to 4v4) and very high relative areas per player (>300 m^2) are used, which is rather impractical in the context of a full-squad environment. With the data from this survey unsuitable to provide clear explanations in this regard, we speculate that the possible reasons may be the following: misconception of HSR and SR definitions, with short-distance acceleration actions, which are very frequent in sided-games,[37] wrongly interpreted as such; use of very low-velocity thresholds (Figure 5), which may mislead and amplify the perception of HSR and SR exposure.

This study has some limitations worth considering. Using a convenience sample, the respondents' sex, roles and professional levels, and their working environment characteristics were not fully and equally represented. Therefore, it is possible that perceptions of other stakeholders involved in decision making (eg, coaches and players) were not considered. Second, the degree to which the data represent other clubs from different confederations/leagues, or athletes of different competition levels and age groups remains uncertain. Finally, being cognizant that current opinions represent only level 5 scientific evidence and higher quality literature on management strategies for HSR and SR exposure in football is somewhat lacking, we acknowledge our findings may change with emerging evidence. Therefore, the perceptions and practices of practitioners should be re-evaluated in the future, based on new research recommendations.

Practical Applications

- Training load management should embed strategies for HSR and SR exposure considering its importance for players' health and performance.
- Standardizing monitoring processes may facilitate data sharing and knowledge exchange between coaching staff, sport science departments, and research groups.
- Synchronous but distinct exposure scheduling and training plans are necessary for starting and nonstarting players across typical and congested weeks.
- Enhancing communication between coaching and sport science staff may help develop or refine ad hoc HSR and SR exposure training solutions.

Conclusions

Building upon the importance of training-load management in football, this study provides actionable insights into the programming constituents of HSR and SR exposure strategies. Universally adopted definitions and methodological procedures are necessary for monitoring exposure systematically and objectively. Training scheduling and implementation should be guided by evidence-based and practice-informed decisions accounting for the contextual factors affecting football performance. While some findings of this study conform with the evidence on HSR and SR training in football, further research and professional debate are warranted to develop empirical knowledge and provide pragmatic recommendations helping practitioners in adopting evidence-based decisions.

Acknowledgment

The authors wish to thank the survey respondents for their participation in this study.

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